The meta-analyses on effectiveness and safety of colorectal cancer screening

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To the Editor: The colorectal cancer (CRC) is the fourth-most common cause of cancer deaths in China that was treated as a public health threat, accounting for approximately 10% of all the cancer-caused mortalities each year.[1] It usually takes decades to develop from a polyp to full CRC,[2] which allows a time window for detecting precancerous lesions and taking early steps for preventions and treatments.

One of the key controls for CRC is screening patients as recommended by the World Health Organization (WHO). The most commonly recommended screening strategies include guaiac-based fecal occult blood testing (gFOBT), fecal immunochemical testing (FIT), colonoscopy, flexible sigmoidoscopy (FS), and computed tomography colonography (CTC). The previously published systematic reviews mainly focused on the studies that were conducted in Europe and America, and most of articles are related to the trends of the general population. However, little scientific evidence on the populations in other regions (e.g., Asia), especially for high-risk populations, were recorded.

In this study, we conducted a systematic review and network meta-analysis (NMA) to assess the evidences for CRC screening from all the regions, and to build a well-developed network to present the evidences from indirect comparisons. This meta analysis followed two predefined protocols that were published on the PROSPERO registry (Nos. CRD42020222440 and CRD42020222441). A number of databases for eligible studies since July 6th, 2020 were searched. The search terms are relevant to CRC and screening strategies. The detailed search strategies for each database can be found in Supplementary File 1, http://links.lww.com/CM9/B983. The snowball retrieval was performed following relevant systematic reviews on CRC screening.

In the study, all the studies including randomized controlled trials (RCTs), cohort studies, quasi-experiments, and case series were considered for inclusion if they met the following criteria: (a) Population: targeting a general population (referring to asymptomatic adults over 18 years old) and/or a high-risk population (including those at the high risks for sporadic CRC and those at the high risk for hereditary CRC, referring to patients with familial adenomatous polyposis, hereditary nonpolyposis CRC, family history of inflammatory bowel disease, colitis, Crohn's disease, and colorectal polyps) for CRC screening. (b) Intervention: application of gFOBT, FIT, colonoscopy, FS, and CTC. (c) Comparisons: no screening measures. (d) Outcomes: effectiveness outcomes (CRC incidence, CRC mortality, and all-cause mortality) and safety outcomes (severe bleeding, perforation, and death caused by screening complications). The definitions of the outcomes are presented in Supplementary File 2, http://links.lww.com/CM9/B983. Only studies in English and Chinese were included.

The following data were extracted: (a) basic reference information; (b) basic information on the study population; (c) interventions: screening strategy, screening interval, and follow-up duration; (d) effectiveness outcomes: the number of deaths from CRC, new patients with CRC, and all-cause deaths; for cohort studies, we

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also extracted the adjusted measures of relative risk, and covariates adjusted; and (e) safety outcomes: the number of severe bleeding events, perforation events, and deaths caused by screening complications.

The Cochrane risk-of-bias tool 2.0 (https://www.riskofbias.info/welcome/rob-2-0-tool) was used to evaluate the quality of the RCT studies, according to diverse RCT study design types. The cohort studies were evaluated using the Newcastle–Ottawa Scale, and the case series studies were evaluated with the tool of the National Institute for Clinical Excellence.

Two groups of independent reviewers (group 1: XYZ, ZXL; group 2: PL, YYL) were selected to be in charge of study selection, information extraction, and quality assessment. If emergency, disagreements and conflicts were resolved by consulting a specialist (FS).

The NMA was performed using data from the RCTs, and a pairwise meta-analysis was conducted with the data from cohort studies since the covariates are diverse. The differences were presented as the 95% confidence interval (CI).

We performed a meta-analysis of single proportions. Studies that reported a proportion of patients who experienced different complications were pooled in three separate meta-analyses. The number of events per 10,000 and the corresponding 95% CIs were calculated. Statistical heterogeneity was assessed using the I^2-value and Cochran's Q statistic. Potential publication bias was assessed using the Egger's test and the funnel plot. Subgroup analysis was conducted according to the age, screening population, and region of study. Sensitivity analysis was performed by excluding studies with a high risk for bias. Statistical analyses were conducted using STATA 14.0 (https://www.stata.com/stata14/) and R 4.1.1. We used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework to rate the quality of evidence for effectiveness.[3]

Overall, 27,423 articles were identified in electronic databases, and 34 articles were manually searched. After screening abstracts and articles, 163 articles were included in our analysis. There were 33 RCTs, 39 cohort studies, and 91 case series in the analysis set. The process of study selection and reasons for exclusion are shown in Supplementary File 3, http://links.lww.com/CM9/B983.

The included studies were conducted in 29 countries, covering North America, Europe, Oceania, and Asia. The total sample sizes of studies varied from 104 to 5,417,699, and the age range of the screened population was above 18 years [Supplementary File 4, http://links.lww.com/CM9/B983]. Most of the included studies were assessed as of low-to-moderate risk of bias [Supplementary File 5, http://links.lww.com/CM9/B983].

Among the 16 RCTs, 9 reported all-cause mortality, 12 reported CRC mortality, and 15 reported CRC incidence. The results of the pairwise meta-analysis are shown in Figure 1A. For CRC mortality, both gFOBT (risk ratio [RR] = 0.86; 95% CI: 0.80–0.94) and FS (RR = 0.76; 95% CI: 0.68–0.85) showed a statistically significant difference compared to the no-screening group. For CRC incidence, only FS (RR = 0.80; 95% CI: 0.74–0.87) showed a statistically significant difference from no-screening. The corresponding forest plots for indirect comparisons are presented in Figure 1B. For CRC mortality, both FS *vs*. gFOBT and FS *vs*. FIT showed no significant differences. For CRC incidence, compared to the gFOBT and FIT, FS (RR = 0.82; 95% CI: 0.73–0.91, RR = 0.83; 95% CI: 0.69–0.99, respectively) showed a lower risk for developing CRC. As for sensitivity analyses, there were no significant changes to our key results, which suggests that the results of our main analysis were robust [Supplementary File 6, http://links.lww.com/CM9/B983]. According to GRADE, the evidence of FS reducing CRC incidence and CRC mortality and gFOBT reducing CRC mortality was assigned to the moderate level. The others were considered low or very low [Supplementary File 7, http://links.lww.com/CM9/B983].

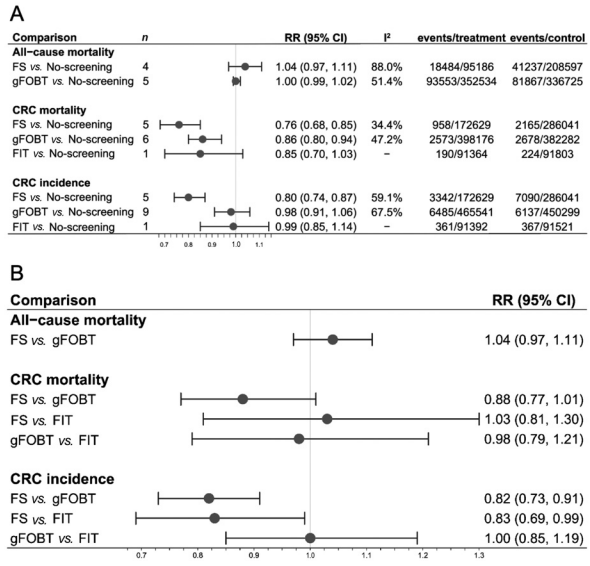


Figure 1: Results of pairwise meta-analysis (A) and indirect meta-analysis (B). CI: Confidence interval; CRC: Colorectal cancer; FIT: Fecal immunochemical test; FS: Flexible sigmoidoscopy; gFOBT: Guaiac-based fecal occult blood test; RR: Risk ratio.

Of all the studies, six were conducted in high-risk populations, and the rest were conducted in general populations. A total of 20 cohorts reported CRC incidence including FIT (n = 6), FOBT (n = 4), colonoscopy (n = 8), and FS (n = 2). As shown in Supplementary File 8, http://links.lww.com/CM9/B983, the pooled results indicated that colonoscopy and FS were associated with a significant reduction of CRC incidence. Colonoscopy was associated with lower CRC incidence in both the general and high-risk populations, particularly in the high-risk population (RR = 0.55 in the general population, 95% CI: 0.54–0.55, I^2 = 8.1%; RR = 0.21 in the high-risk population, 95% CI: 0.18–0.25, I^2 = 0%). A total of 16 cohorts reported CRC mortality including colonoscopy (n = 7), FIT (n = 4), gFOBT (n = 4), and FS (n = 2); and one cohort[4] compared FS and colonoscopy with an unscreened group. The pooled results showed that in the general population,

compared to the unscreened procedures, FIT (RR = 0.50; 95% CI: 0.34–0.75; I^2 = 46.7%), colonoscopy (RR = 0.36; 95% CI: 0.23–0.56; I^2 = 74.4%), and gFOBT (RR = 0.66; 95% CI: 0.45–0.95; I^2 = 98.7%) significantly reduced CRC mortality. Two cohorts where FS was compared to non-screening showed a significant difference (RR = 0.59; 95% CI: 0.46–0.95; I^2 = 0%). The details are given in Supplementary File 8, http://links.lww.com/CM9/B983. A total of six cohorts reported all-cause mortality, including FIT (n = 1), colonoscopy (n = 3), gFOBT (n = 1), and FS (n = 1). All three cohorts undergoing colonoscopy were from a high-risk population, indicating that colonoscopy could reduce all-cause mortality by 77% [Supplementary File 8, http://links.lww.com/CM9/B983]. Detailed results of pairwise meta-analysis for cohort studies are shown at the individual study level in Supplementary File 9, http://links.lww.com/CM9/B983.

There were 108 groups in 97 studies that reported severe bleeding caused by screening, 131 groups in 113 studies that reported perforation caused by screening, and 59 groups in 53 studies that reported death caused by screening complications [Supplementary File 10, http://links.lww.com/CM9/B983]. Pooled rates of severe bleeding, perforation, and death caused by complications after colonoscopy, CTC, and FS are shown in Supplementary File 11, http://links.lww.com/CM9/B983.

Subgroup analyses were conducted according to screening population, study region, and age (>50 years) [Supplementary File 12, http://links.lww.com/CM9/B983]. The severe bleeding rate caused by colonoscopy was lower in Asia than that in Europe, North America, and Australia. The perforation rate caused by CTC was higher in Asia than that in Europe, North America, and Australia. Other complication rates in Asia were comparable to those in Europe, North America, and Australia.

Our comprehensive systematic review and NMA of RCTs and observational studies found that FS and colonoscopy were effective screening strategies for preventing CRC, as they decreased the incidence and mortality of CRC. FIT and gFOBT did not have significantly different effectiveness. Although GRADE scores were low to moderate for most comparisons, FS seems to be an advisable option for CRC screening. Colonoscopy is associated with a notable risk of complications, while FS appears to have a relatively low risk of complications. CTC is considered safe, but its effectiveness has yet to be evaluated.

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Conflicts of interest

None.

Data sharing statement

All data generated or analyzed during this study are included in the article (and/or its supplementary).

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